Requirements table

Lab 1

Blood bank inventory & emergency donor matcher

Team number:12

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Req ID	Type	Description	Priority	Acceptance criteria	Rationale
FR-001	Functional	The system shall cross-reference emergency blood unit requests against real-time inventory and broadcast SMS alerts to compatible donors within a 10 km radius.	High	Pass: compatible donors notified within 30 seconds of the emergency flag. Fail: an incompatible blood group is alerted.	Core emergency-response function of the system.
FR-002	Functional	The system shall allow the blood bank manager to add and update blood units, recording blood type, quantity, collection date, and expiry date.	High	Pass: a newly logged unit appears in the inventory list within 5 seconds with all four fields populated. Fail: a unit is saved with a missing field.	Accurate stock records are required before any matching or alerting can happen.
FR-003	Functional	The system shall automatically flag blood units within 3 days of expiry and remove expired units from the available stock count.	High	Pass: a unit reaching its expiry date is excluded from available stock within 1 hour. Fail: an expired unit is still offered as available.	Prevents unsafe or wasted stock from being allocated during emergencies.
FR-004	Functional	The system shall allow a donor to register or update their blood type and location for eligibility matching.	Medium	Pass: a donor's profile update is reflected in matching results within 1 minute. Fail: an outdated blood type is used in a match.	Accurate donor profiles are required for correct compatibility matching.
FR-005	Functional	The system shall allow a notified donor to accept or decline an emergency alert, and shall log the response.	Medium	Pass: a donor's accept/decline response is recorded with a timestamp within 10 seconds of submission. Fail: a response is not logged.	Enables the blood bank to track real donor availability, not just notification delivery.

NFR-001	Nonfunctional — performance & security	The inventory ledger must maintain strict transactional consistency, preventing simultaneous allocation of the same blood bag to two requests.	High	Pass: benchmarking tests under simulated peak load confirm target latency with zero double-allocations. Fail: any double-allocation occurs.	Prevents a critical safety failure where one unit is promised to two patients.
NFR-002	Nonfunctional — privacy & security	Donor location data must be encrypted in transit and at rest, and used only for radius-based matching.	High	Pass: a security audit confirms encryption at rest and in transit, and confirms location data is not retained beyond the matching window. Fail: any unencrypted storage or transmission is found.	Protects sensitive donor location data from exposure or misuse.